

# **Minimal-Shot AV Evidence Deck**

## **SOTA Submission Results + CoRL Transfer Diagnostics**

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Waymo Open Dataset E2E Challenge / SOTA + CoRL 2027 branch

# Two Audiences, One Evidence Package

This deck supports two related but distinct uses.

| Audience   | Emphasis                                   | Boundary             |
|------------|--------------------------------------------|----------------------|
| SOTA audit | working system, bundles, WOD validation-CV | not production       |
| CoRL paper | transfer failure diagnosis                 | not method dominance |

Do not collapse them into one claim.

The SOTA submission is a system/evidence package. The CoRL story is a mechanistic diagnosis built on that infrastructure.

# SOTA Submission Scope

Primary SOTA track:

- episode-free Spotlight Reflex closed-loop runtime
- randomized WOD-style long-tail scenarios
- explicit minimal-shot integrity audit
- no public-road deployment or legal certification claim

Auxiliary benchmark track:

- WOD-E2E candidate-selection harness
- segment-grouped validation-CV only
- frozen Cosmos / InternVLA grounding ablations

Source: `artifacts/sota_judging_criteria_audit.json`,

# SOTA Readiness Gates

| Gate                                     | Status |
|------------------------------------------|--------|
| Grand archive ready                      | pass   |
| Minor archive ready                      | pass   |
| Minimal-shot integrity                   | pass   |
| Judging-criteria evidence                | pass   |
| Production no-go boundary declared       | pass   |
| WOD symbolic-trust risk-control boundary | pass   |

No blockers are reported in `artifacts/final_submission_readiness_audit.json`.

The submission boundary is explicit:

`submission_candidate_not_hidden_test_not_production`.

# SOTA Primary Simulation Result

Closed-loop simulation headline from the existing SOTA evidence package:

| Suite                      | Runs | Collisions | Pass  |
|----------------------------|------|------------|-------|
| WOD-style, all 11 clusters | 110  | 0          | 100%  |
| Compositional OOD          | 60   | 0          | 100%  |
| Adversarial                | 60   | 0          | 100%  |
| Hidden holdout             | 60   | 0          | 100%  |
| Gauntlet                   | 60   | 0          | 60%   |
| Total                      | 350  | 0          | 93.1% |

COMPASS: **9.137 / 10** over 700 ranked runs.

Source: [README.md](#) | [artifacts/compass\\_evidence\\_report.json](#)

# SOTA Baseline Comparison

Matched gauntlet comparison: same 420 scenarios, same seeds.

| Policy                             | Pass rate    | Collision rate |
|------------------------------------|--------------|----------------|
| Baseline, no world-state reasoning | 2.1%         | 20.5%          |
| Spotlight Reflex                   | <b>57.6%</b> | <b>7.9%</b>    |

Interpretation for SOTA:

- the system contribution is not just a score
- it is an auditable geometry-grounded runtime path
- failures are analyzed rather than hidden behind an aggregate metric

Source: `README.md` , `docs/presentation.tex` .

# WOD-E2E Auxiliary Benchmark

479 WOD-E2E validation frames, segment-grouped CV.

| Selector             | RFS          | Folds | Role                   |
|----------------------|--------------|-------|------------------------|
| Waymo baseline       | 7.022        | -     | reference              |
| Local baseline       | 7.131        | -     | reference              |
| Gate-only            | 7.803        | 5     | selector baseline      |
| RFF direct policy    | 7.834        | 5     | learned baseline       |
| GPU MLP + Cosmos 64d | <b>7.845</b> | 5     | champion validation-CV |
| Oracle selector      | 9.264        | 5     | upper bound            |

This is auxiliary evidence, not a hidden-test or zero-shot WOD claim.

Source: README.md, artifacts/used\_grounding\_ablation\_table.md

# CoRL Research Claim

Internal closed-loop success is not a reliable transfer proxy.

We find that simulator transfer failure decomposes into separate axes:

- geometry: offroad and route reconstruction
- safety: collision and proximity violation
- progress: distance and route completion
- lane adherence: wrong-lane or lane-violation behavior

No evaluated intervention is co-monotone across all axes in AlpaSim.



# Why This Matters Now

Recent VLA work moves toward physically grounded latent reasoning.

LaST-VLA argues that autonomous-driving VLA reasoning should use 3D geometric priors

and world-model dynamics, not only text chain-of-thought.

It reports NAVSIM v1/v2 gains through latent spatio-temporal reasoning.

Our question is narrower and diagnostic:

If a policy or selector is internally strong, which transfer axes actually survive when the observation and dynamics interface changes?

Reference: [LaST-VLA, arXiv:2603.01928](#)

# Evidence Stack

| Layer                   | Purpose                                         | Evidence                                      |
|-------------------------|-------------------------------------------------|-----------------------------------------------|
| SOTA runtime            | Working minimal-shot AV system                  | Spotlight Reflex, COMPASS, submission bundles |
| Custom simulator        | Controlled long-tail AV stress lab              | LHS generator, DAgger, proxy perturbations    |
| AlpaSim adapter         | External sim-to-sim validation on WOD-E2E clips | 10 matched clips, partial 13-scene refresh    |
| WOD candidate-selection | Real-data grounding probe                       | 479 validation frames, segment-grouped CV     |

The tracks are deliberately separated.

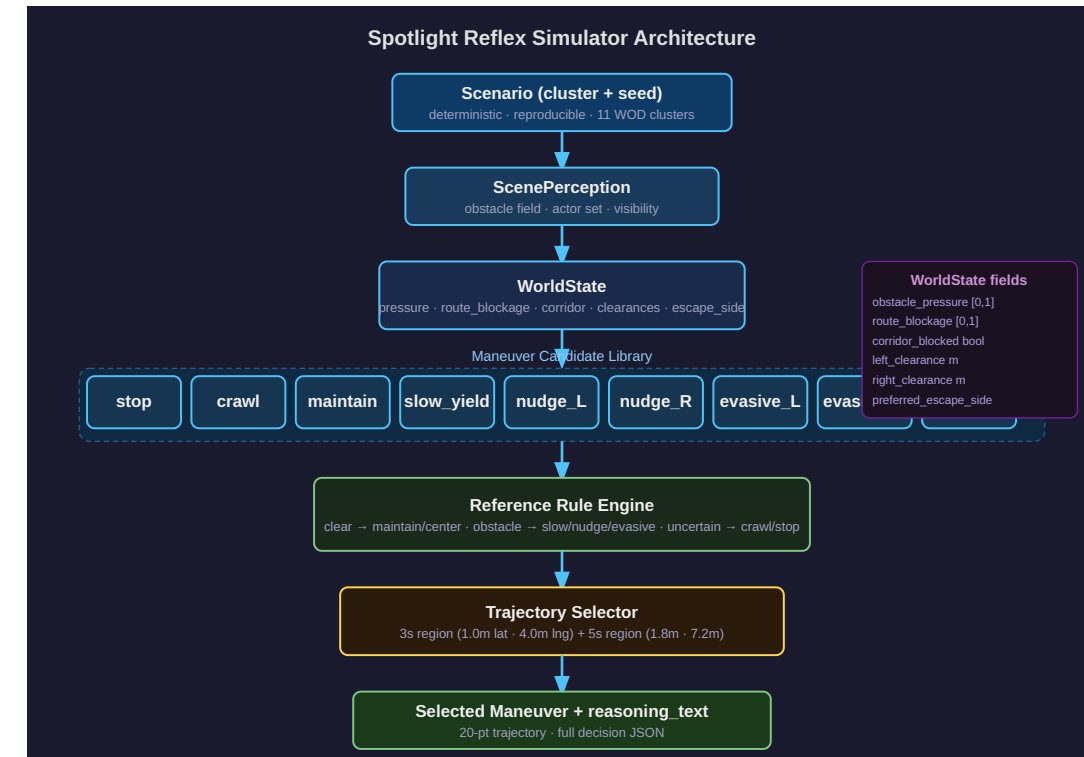
Simulator metrics are not used to tune WOD selector results and WOD validation

# What the Simulator Tests

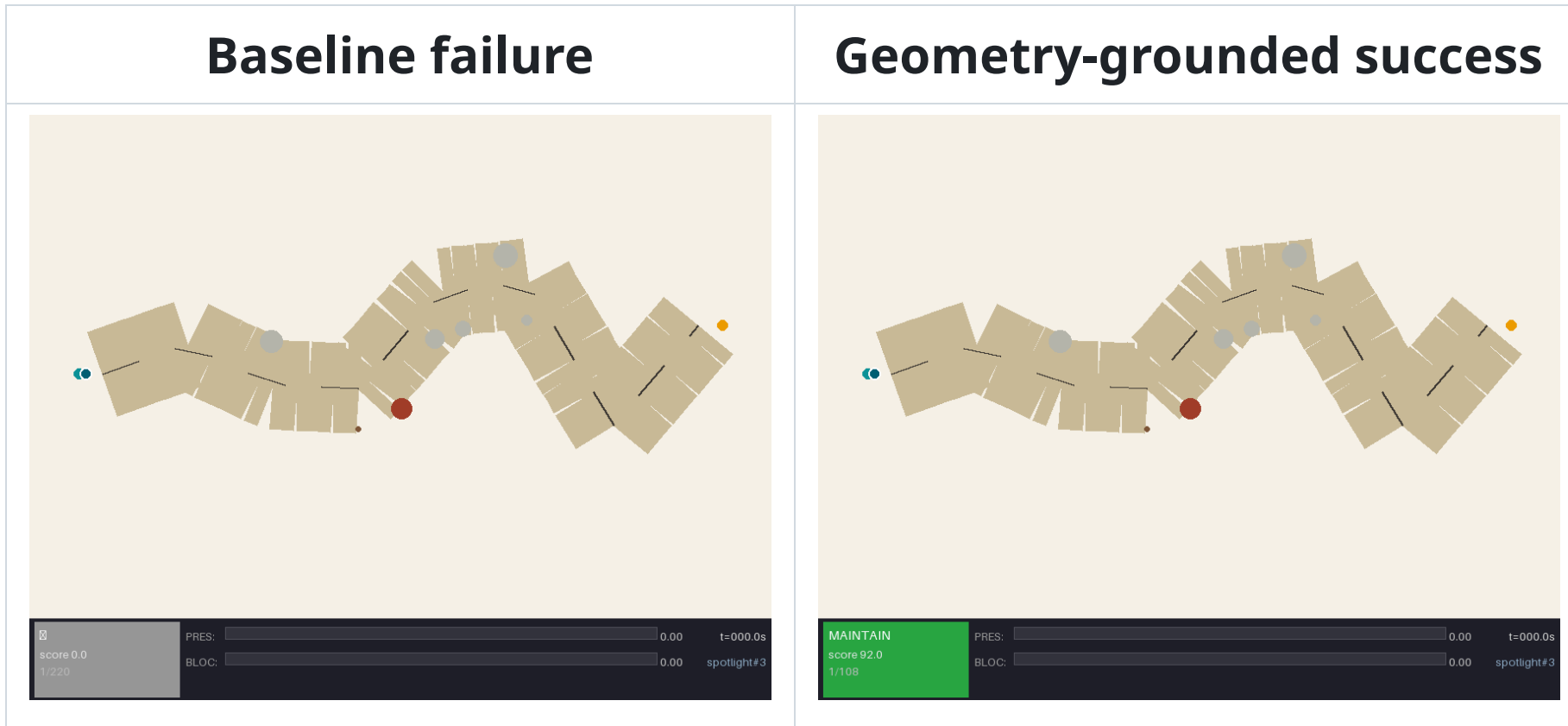
The internal simulator is a controlled micro-lab, not a realism claim.

It lets us isolate:

- discrete token action spaces
- rule-based geometric selection
- BC and DAgger imitation variants
- proxy-state corruption with true-state evaluation
- matched seeds across policies and perturbations



# Visual Failure and Success Cases



Same scenario family: wrong-way actor under low visibility.

The baseline extrapolates into the hazard. Spotlight selects an evasive token from

# Policy Interface

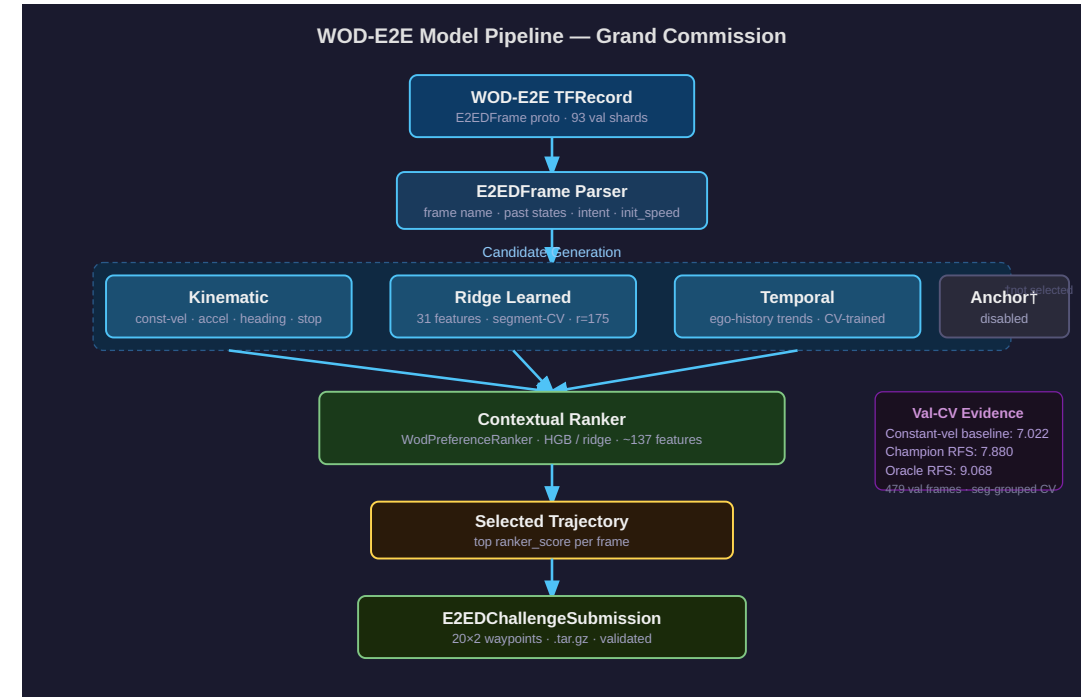
State:

- 10 scalar geometric features for learned policies
- six core world-state scalars for Spotlight

Action:

- 9 ManeuverTokens
- stop, crawl, maintain, slow-yield
- nudge left/right, evasive left/right, lane-recover

The policy selects a token; the



# Internal Result: DAgger Closes the Local Frontier

Held-out Latin-hypercube sweep: 12 profiles, seeds 1-10, Gauntlet / Adversarial / Hidden.

Each agent: 1080 closed-loop rollouts.

| Agent         | Overall pass / PV | Gauntlet     | Adversarial  | Hidden       |
|---------------|-------------------|--------------|--------------|--------------|
| Continuous-BC | 77.69 / 3.52      | 77.50 / 4.31 | 75.42 / 2.08 | 83.33 / 1.67 |
| Token-BC      | 76.85 / 4.26      | 77.64 / 4.86 | 71.67 / 2.92 | 82.50 / 3.33 |
|               |                   | 79.17 /      |              | 81.67 /      |

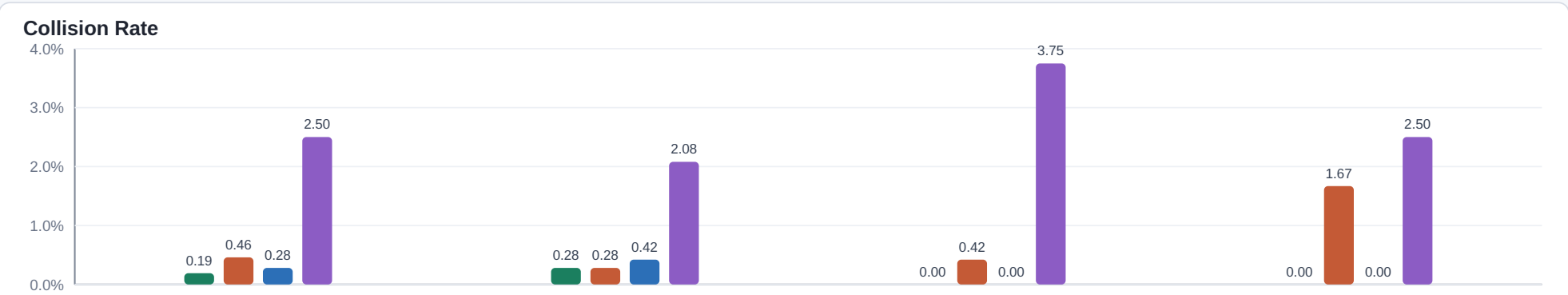
# Internal Result: More DAgger Is Not Monotonic

| Policy               | Overall pass / PV   | Gauntlet            | Adversarial         | Hidden              |
|----------------------|---------------------|---------------------|---------------------|---------------------|
| Token-DAgger, 1-step | 93.98 / 0.46        | 96.81 / 0.56        | 87.50 / 0.42        | 90.00 / 0.00        |
| Token-DAgger, 2-step | <b>94.54 / 0.19</b> | <b>96.25 / 0.28</b> | <b>90.83 / 0.00</b> | <b>91.67 / 0.00</b> |
| Token-DAgger, 3-step | 92.50 / 0.46        | 94.31 / 0.28        | 87.92 / 0.42        | 90.83 / 1.67        |
| 3-step source decay  | 93.06 / 0.28        | 94.86 / 0.42        | 89.17 / 0.00        | 90.00 / 0.00        |
| 3-step no inverse    | 89.35 / 2.50        | 92.08 /             | 82.50 / 3.75        | 86.67 /             |

# Dagger Aggregation Visual

## Dagger Aggregation Ablations on Held-Out Latin-Hypercube Sweep

Two-step Dagger remains the frontier. Source decay partially recovers the three-step collapse; removing inverse-frequency weights breaks closed-loop safety.





# Negative Result: Scoring Ego Trajectories Was Not Enough

We trained trajectory-informed scorers over candidate futures with:

- curvature and heading deltas
- path length, acceleration, jerk proxies
- minimum dynamic clearance and time-to-closest-approach
- signed lateral and longitudinal miss distances

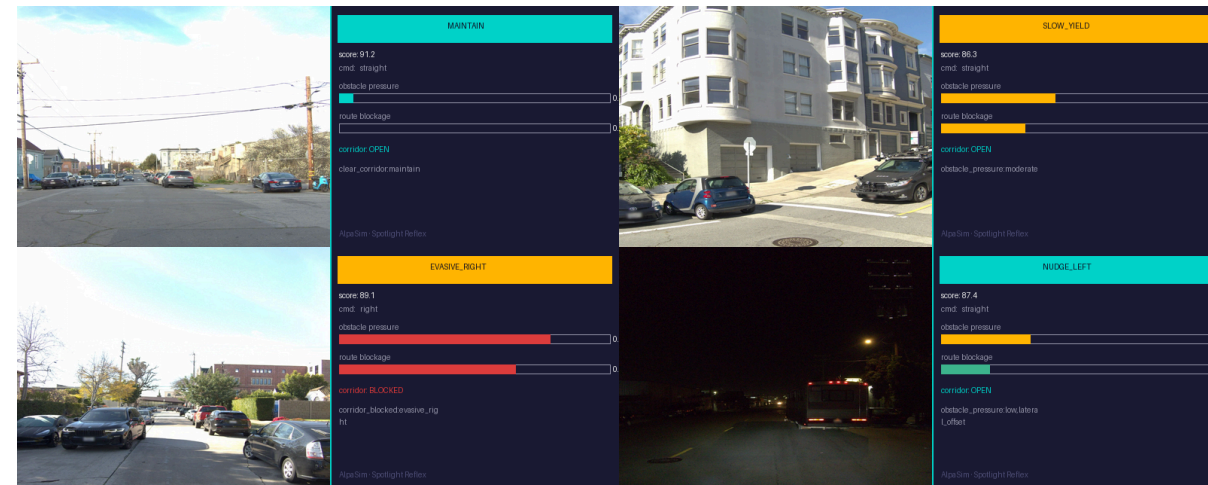
| Policy                                   | Gauntlet pass / PV | Adversarial  | Hidden       |
|------------------------------------------|--------------------|--------------|--------------|
| Interaction scorer, learner states       | 86.81 / 1.94       | 85.00 / 1.67 | 89.17 / 1.67 |
| Interaction scorer, learner states + ... | 86.81 / 1.94       | 85.00 / 1.67 | 89.17 / 1.67 |

# External Diagnostic: AlpaSim

AlpaSim uses real WOD-E2E front-camera frames and an adapter that reconstructs:

- brightness
- route command
- ego dynamics
- sparse hazard primitives
- geometric scalars consumed by the policy

This is a transfer diagnostic, not a



# AlpaSim Bridge

The external bridge keeps the policy binary fixed.

What changes is the interface:

- camera and route observations become proxy geometric scalars
- sparse hazards become obstacle and actor primitives
- token selection is evaluated under AlpaSim execution

This is exactly where sim-to-sim mismatch is exposed.

# AlpaSim Matrix: One Fix Does Not Repair All Axes

10 matched WOD-E2E clips completed by all four variants.

| <b>Model</b>        | <b>Collision</b> | <b>Offroad</b> | <b>Wrong<br/>lane</b> | <b>Progress</b> | <b>Dist.<br/>m</b> | <b>Dist.-<br/>GT</b> |
|---------------------|------------------|----------------|-----------------------|-----------------|--------------------|----------------------|
| Raw DAgger<br>iter2 | 0.600            | 0.900          | 0.700                 | 0.034           | 61.98              | 13.86                |
| Clamped lateral     | 0.700            | 0.500          | 0.200                 | 0.384           | 59.85              | 5.21                 |
| Hard veto<br>hybrid | 0.800            | 0.200          | 0.700                 | 0.837           | 166.06             | 22.71                |
| Source decay        | 0.600            | 0.900          | 0.400                 | 0.175           | 62.92              | 27.16                |

# What the AlpaSim Interventions Show

| Intervention     | Repairs                                       | Worsens or fails                          |
|------------------|-----------------------------------------------|-------------------------------------------|
| Clamped lateral  | offroad, wrong-lane, progress, distance-to-GT | collision increases                       |
| Hard veto hybrid | progress, offroad                             | collision increases, wrong-lane unchanged |
| Source decay     | wrong-lane, progress                          | offroad unchanged, distance-to-GT worsens |

The hard veto is especially diagnostic:

- vetoed 1987 / 1990 DAgger argmax decisions
- selected mostly the geometric scorer path

# Controlled Proxy-State Perturbation

Internal simulator, true-state physics and evaluation fixed.

Only the planner-visible proxy state is corrupted.

| Perturbation  | Raw p/c/l         | Clamp             | Hybrid                   |        |
|---------------|-------------------|-------------------|--------------------------|--------|
| clean         | 0.917/0.083/0.312 | 0.938/0.062/0.167 | 0.958/0.042/0.167        | 0.938, |
| heading bias  | 0.896/0.104/0.562 | 0.938/0.062/0.396 | 0.938/0.062/0.375        | 0.896, |
| actor latency | 0.604/0.396/0.875 | 0.604/0.396/0.521 | <b>0.854/0.146/0.208</b> | 0.833, |
| route offset  | 0.938/0.062/0.229 | 0.875/0.125/0.021 | 0.896/0.104/0.062        | 0.938, |

Source: `artifacts/internal_proxy_transfer_medium.md`.

# Why the Proxy Sweep Matters

AlpaSim alone leaves an adapter-confound objection.

The proxy sweep removes that escape hatch:

- same simulator dynamics
- same token library
- same controller
- matched scenarios
- only proxy state is corrupted

The same multi-axis tradeoff appears internally under controlled corruption.

That makes the external result mechanistic rather than anecdotal.

# The Proposition

Let each candidate token have a metric vector:

```
m(a) = [collision risk, offroad risk, lane risk, progress loss, tracking error]
```

If a proxy-state perturbation changes the action ordering for one metric but not another,

a scalar intervention can improve one axis while worsening another.

The empirical question is not whether this is possible.

The empirical question is whether it occurs repeatedly under realistic AV transfer interfaces.

Our answer: yes, in AlpaSim and in controlled proxy perturbations.



# WOD Grounding Probe

479 WOD-E2E validation frames, segment-grouped CV.

| Ablation                         | RFS          | Oracle | Regret | Oracle match | Grounding signal        |
|----------------------------------|--------------|--------|--------|--------------|-------------------------|
| Scalar / geometry only           | 7.695        | 9.068  | 1.373  | 0.403        | none                    |
| InternVLA only                   | 7.672        | 9.225  | 1.554  | 0.392        | InternVLA               |
| Cosmos + InternVLA linear fusion | 7.728        | 9.208  | 1.480  | 0.403        | Cosmos + InternVLA      |
| Cosmos 64d nonlinear head        | <b>7.845</b> | 9.264  | 1.419  | 0.390        | Cosmos + nonlinear head |

Grounding helps only with the right head

# CoRL Evidence Strength Audit

Current CoRL audit conclusion: `strong_diagnostic` .

| Gate                         | Current                    | Target           | Status  |
|------------------------------|----------------------------|------------------|---------|
| Internal proxy diagnostic    | 288 cases                  | $\geq 288$ cases | pass    |
| External transfer diagnostic | 10 matched scenes          | $\geq 10$ scenes | pass    |
| Larger external scale        | 10 scenes                  | $\geq 30$ scenes | not yet |
| Uniform positive method      | 0 external dominance cases | $\geq 1$         | not yet |
| Grounding prior signal       | $\pm 0.150$ RFS            | $\geq 0$         | pass    |

# SOTA vs CoRL Claim Boundary

| Question                              | Answer                                            |
|---------------------------------------|---------------------------------------------------|
| Is the SOTA submission packaged?      | yes, readiness audit has no blockers              |
| Is this a production AV safety claim? | no, production audit is no_go                     |
| Is WOD 7.845 a hidden-test claim?     | no, validation-CV only                            |
| Is the CoRL method a uniform winner?  | no, external dominance count is 0                 |
| Is the CoRL diagnosis supported?      | yes, internal proxy + AlpaSim both show tradeoffs |

This boundary should stay visible in the presentation.

# Media and Figures Included

| Asset                                                    | Use                         |
|----------------------------------------------------------|-----------------------------|
| <code>docs/images/baseline_spotlight.gif</code>          | baseline visual failure     |
| <code>docs/images/spotlight_success.gif</code>           | geometry-grounded success   |
| <code>docs/images/construction_success.gif</code>        | long-tail construction case |
| <code>docs/images/intersection_stress.gif</code>         | multi-actor stress case     |
| <code>docs/images/alpasim_reasoning_panel.png</code>     | AlpaSim adapter reasoning   |
| <code>docs/images/dagger_aggregation_ablation.svg</code> | Dagger aggregation plot     |
| <code>docs/images/alpasim-bridge.svg</code>              | external bridge diagram     |

GIFs are used as short videos in GitHub, Marp, and most slide renderers.

# Additional Visual Cases



These examples are not the proof.

They show what the controlled simulator is stressing: narrow corridors, route

# Reproducibility Path for Auditors

SOTA submission audits:

```
artifacts/final_submission_readiness_audit.json  
artifacts/sota_judging_criteria_audit.json  
artifacts/minimal_shot_claim_audit.json  
artifacts/sota_submission_bundles/
```

CoRL audit wrapper:

```
./scripts/run_corl2027_audit.sh
```

Core evidence files:

```
docs/corl2027/paper.tex  
docs/corl2027/AUDIT.md  
artifacts/corl_evidence_strength_audit.md  
artifacts/internal_proxy_transfer_medium.md
```

# Submission Framing

For SOTA, frame this as:

“A reproducible minimal-shot AV system with explicit claim boundaries, simulation evidence,

WOD validation-CV auxiliary results, and packaged audit artifacts.”

For CoRL, do not frame this as:

“Our hybrid method dominates.”

The data does not support that.

Frame the CoRL result as:

“Internal imitation-learning success can hide transfer-axis disagreement.

We provide a controlled diagnostic, reproduce the effect externally in AlpaSim,

# Takeaway

The surprising result is not that DAgger fails externally.

The result is where it fails:

- the internal simulator says the learned token policy is calibrated
- AlpaSim shows the selected tokens are often externally incompatible
- controlled proxy corruption reproduces the same axis conflicts
- frozen world/VLA priors help only through head-sensitive grounded selection

Robust AV evaluation needs per-axis transfer diagnostics, not only offline accuracy or aggregate closed-loop pass rate.